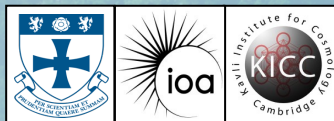


Does the Model Fit the Data?

*Machine learning tools to detect
gravitational wave mismodelling
(Summer '25 @ the IoA)*



T G Hardy

7th Nov @ IoA 30yr Alumni

Supervised by Dr James Alvey

KICC/IoA @ Cambridge University



ANNALES DE L'OBSERVATOIRE IMPÉRIAL DE PARIS.

RECHERCHES ASTRONOMIQUES (SUITE) [*].

PAR U.-J. LE VERRIER.

CHAPITRE XV.

THÉORIE DU MOUVEMENT DE MERCURE.

Adeo ut coelestis hic Mercurius non minus Astronomos
torserit, quam terrestris Alchimistas eludat.

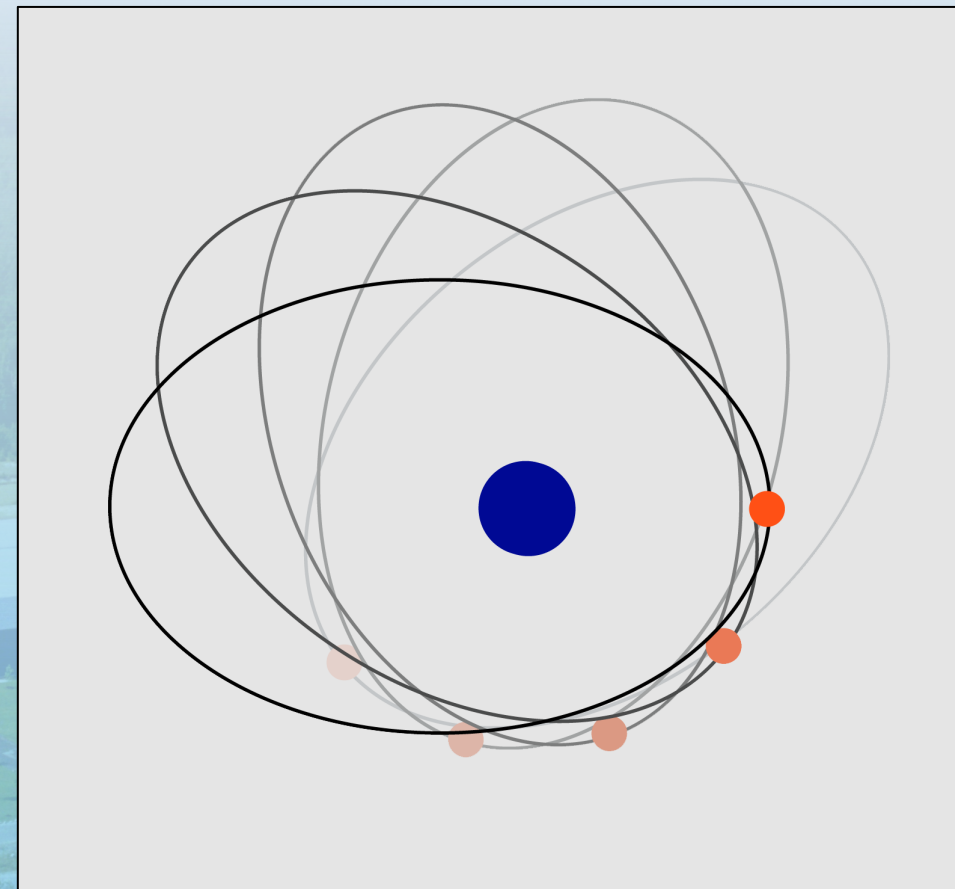
(RICCIOLI, *Almagest. nov.* Lib. VII, sect. III, cap. I.)

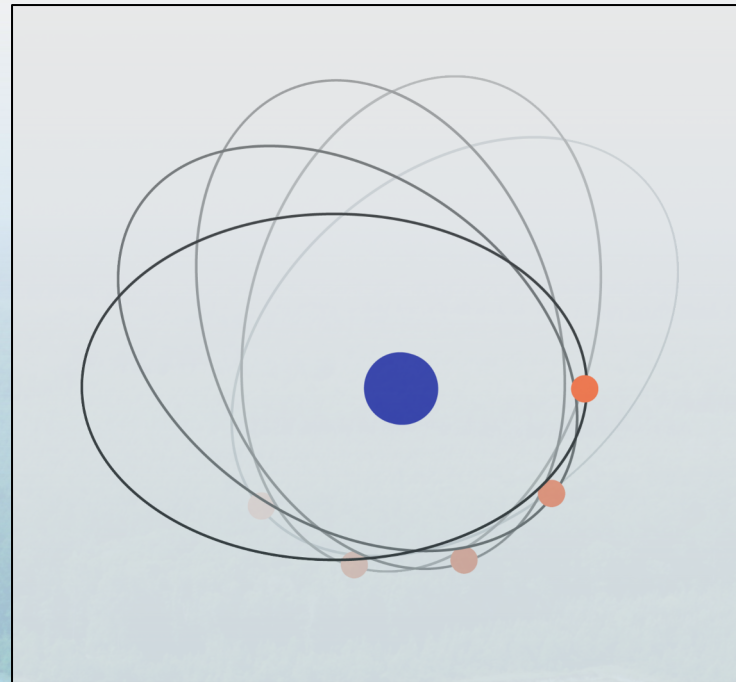
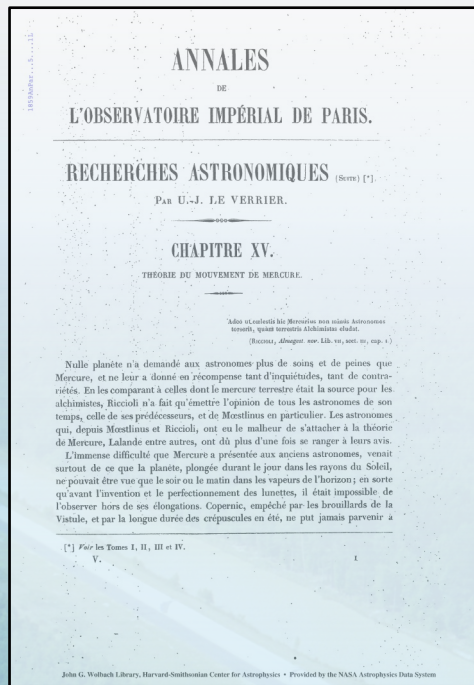
Nulle planète n'a demandé aux astronomes plus de soins et de peines que Mercure, et ne leur a donné en récompense tant d'inquiétudes, tant de contrariétés. En les comparant à celles dont le mercure terrestre était la source pour les alchimistes, Riccioli n'a fait qu'émettre l'opinion de tous les astronomes de son temps, celle de ses prédécesseurs, et de Mœstlinus en particulier. Les astronomes qui, depuis Mœstlinus et Riccioli, ont eu le malheur de s'attacher à la théorie de Mercure, Lalande entre autres, ont dû plus d'une fois se ranger à leurs avis.

L'immense difficulté que Mercure a présentée aux anciens astronomes, venait surtout de ce que la planète, plongée durant le jour dans les rayons du Soleil, ne pouvait être vue que le soir ou le matin dans les vapeurs de l'horizon ; en sorte qu'avant l'invention et le perfectionnement des lunettes, il était impossible de l'observer hors de ses elongations. Copernic, empêché par les brouillards de la Vistule, et par la longue durée des crépuscules en été, ne put jamais parvenir à

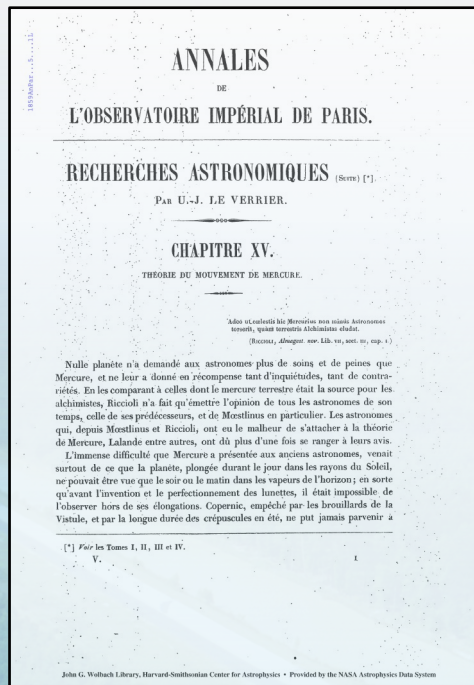
[*] Voir les Tomes I, II, III et IV.

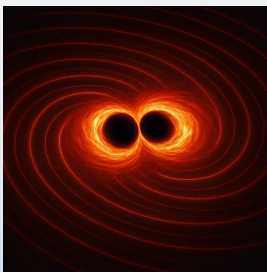
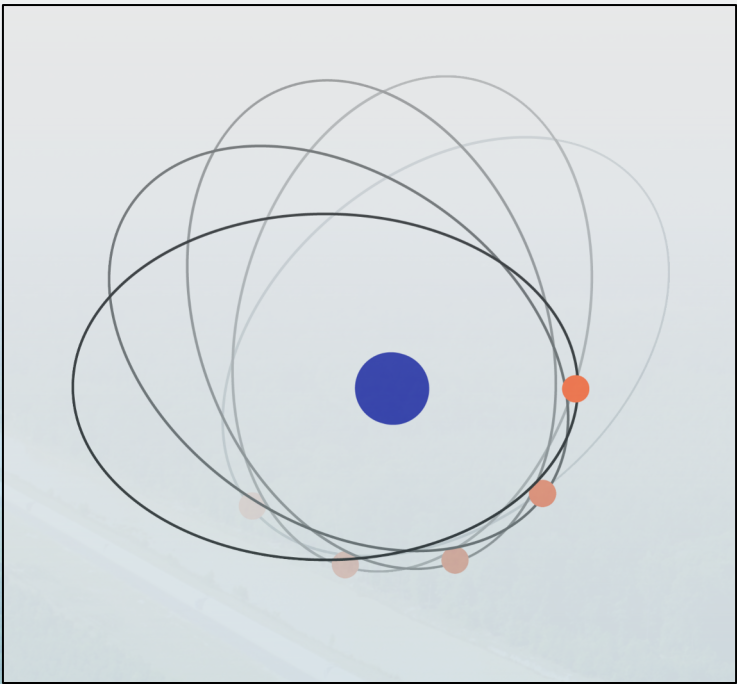
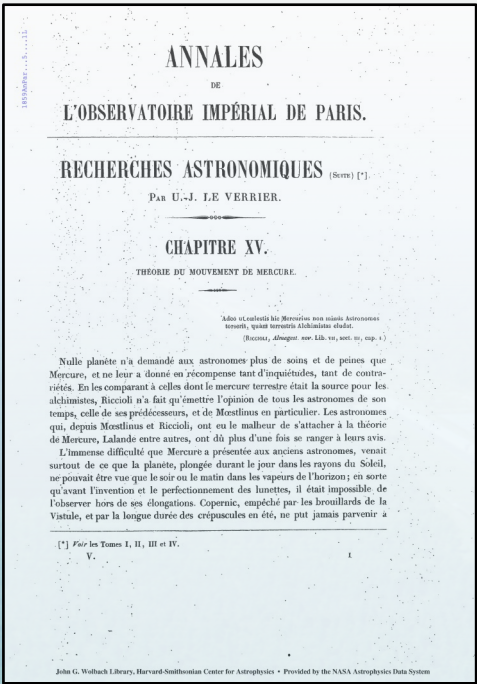
Zenz (2007)





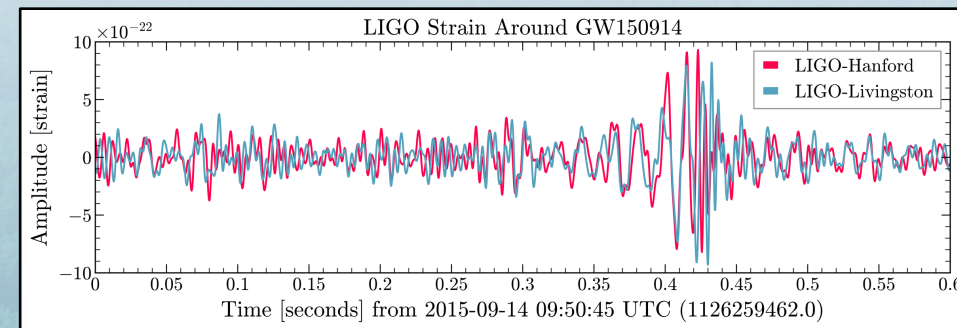
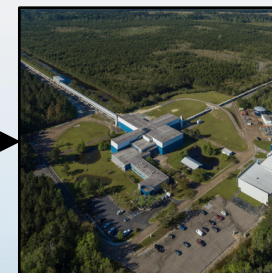
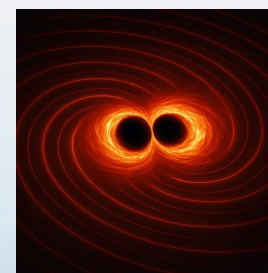
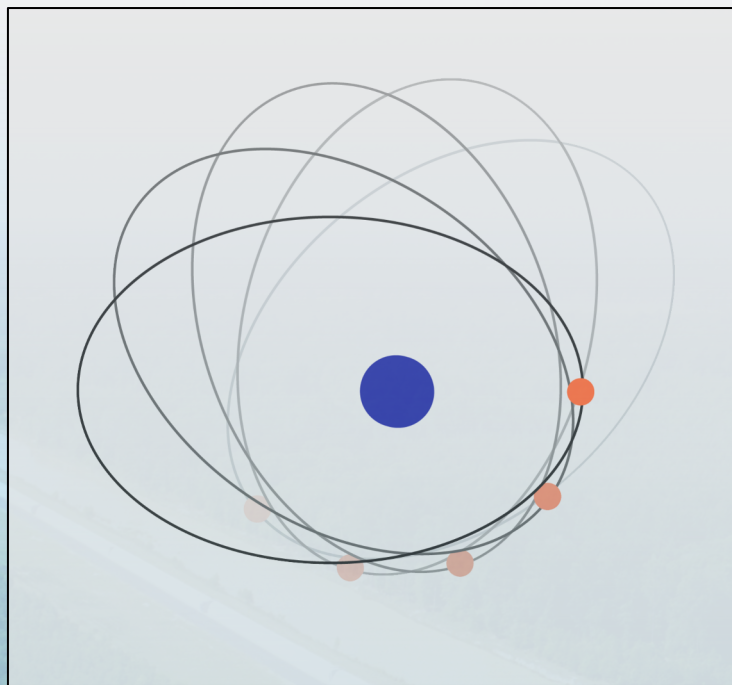
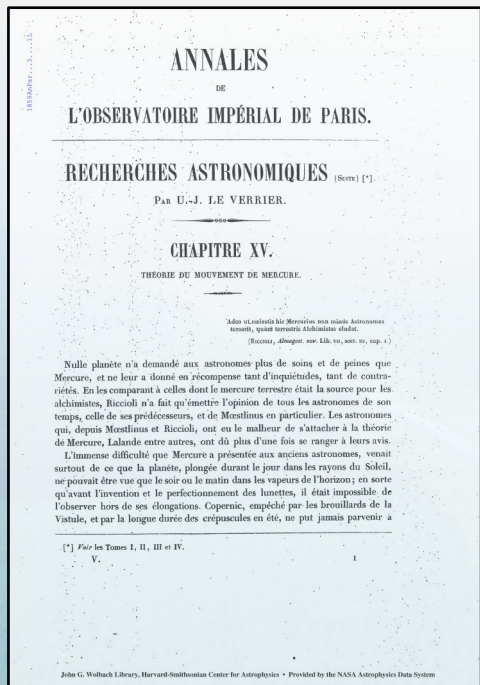
$$P^2 = \frac{4\pi^2}{GM} a^3$$





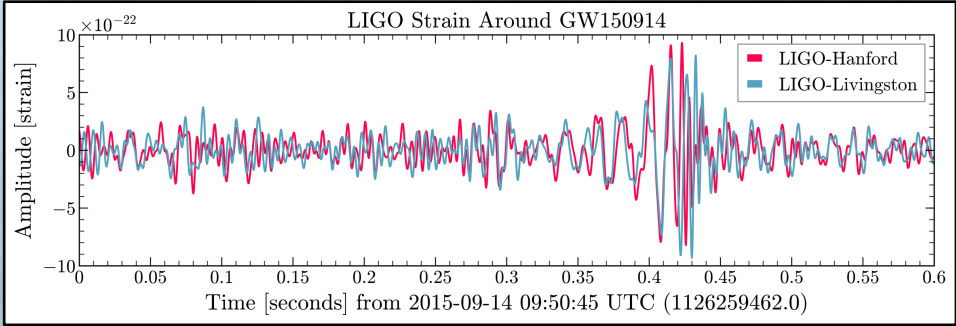
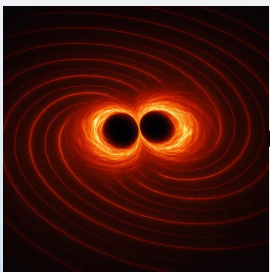
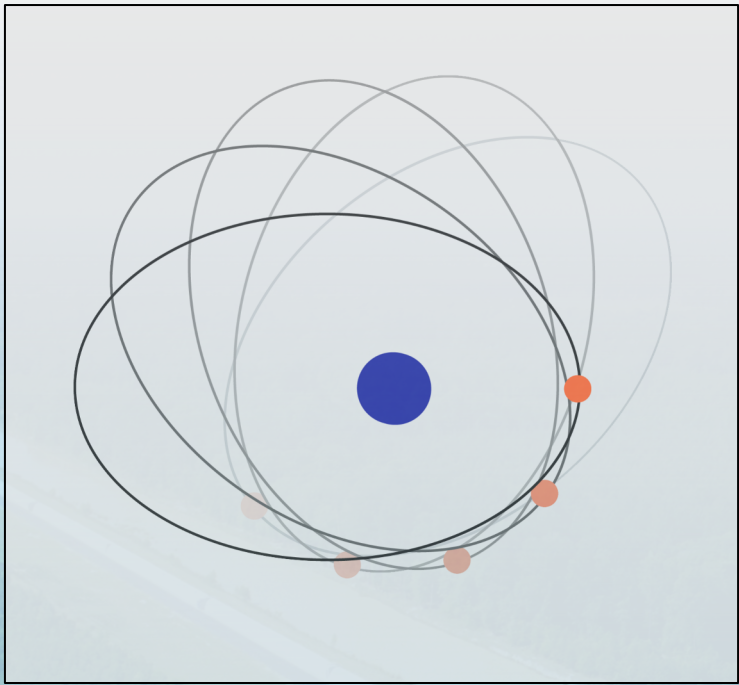
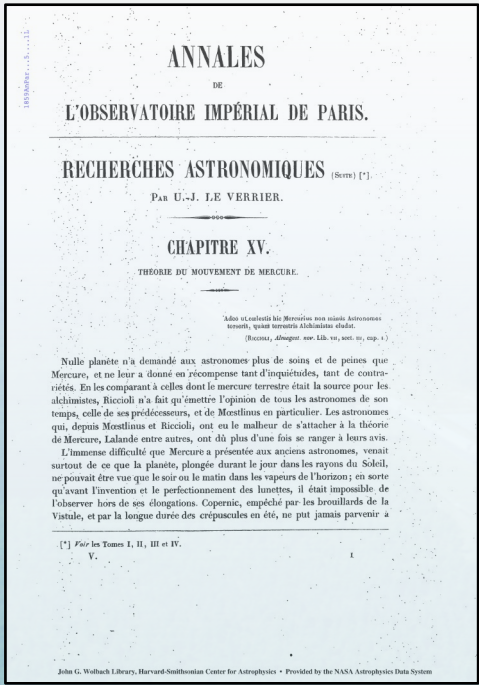
$$P^2 = \frac{4\pi^2}{GM} a^3$$

$$\Delta\phi \approx \frac{6\pi GM}{c^2 a (1 - e^2)}$$



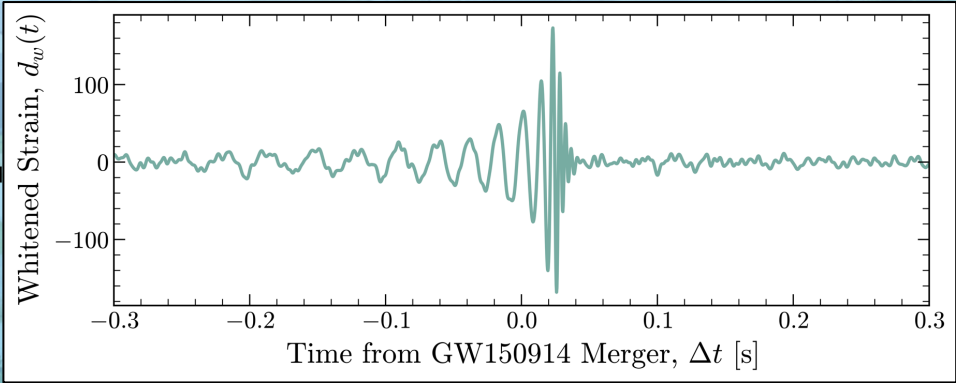
$$P^2 = \frac{4\pi^2}{GM} a^3$$

$$\Delta\phi \approx \frac{6\pi GM}{c^2 a(1 - e^2)}$$



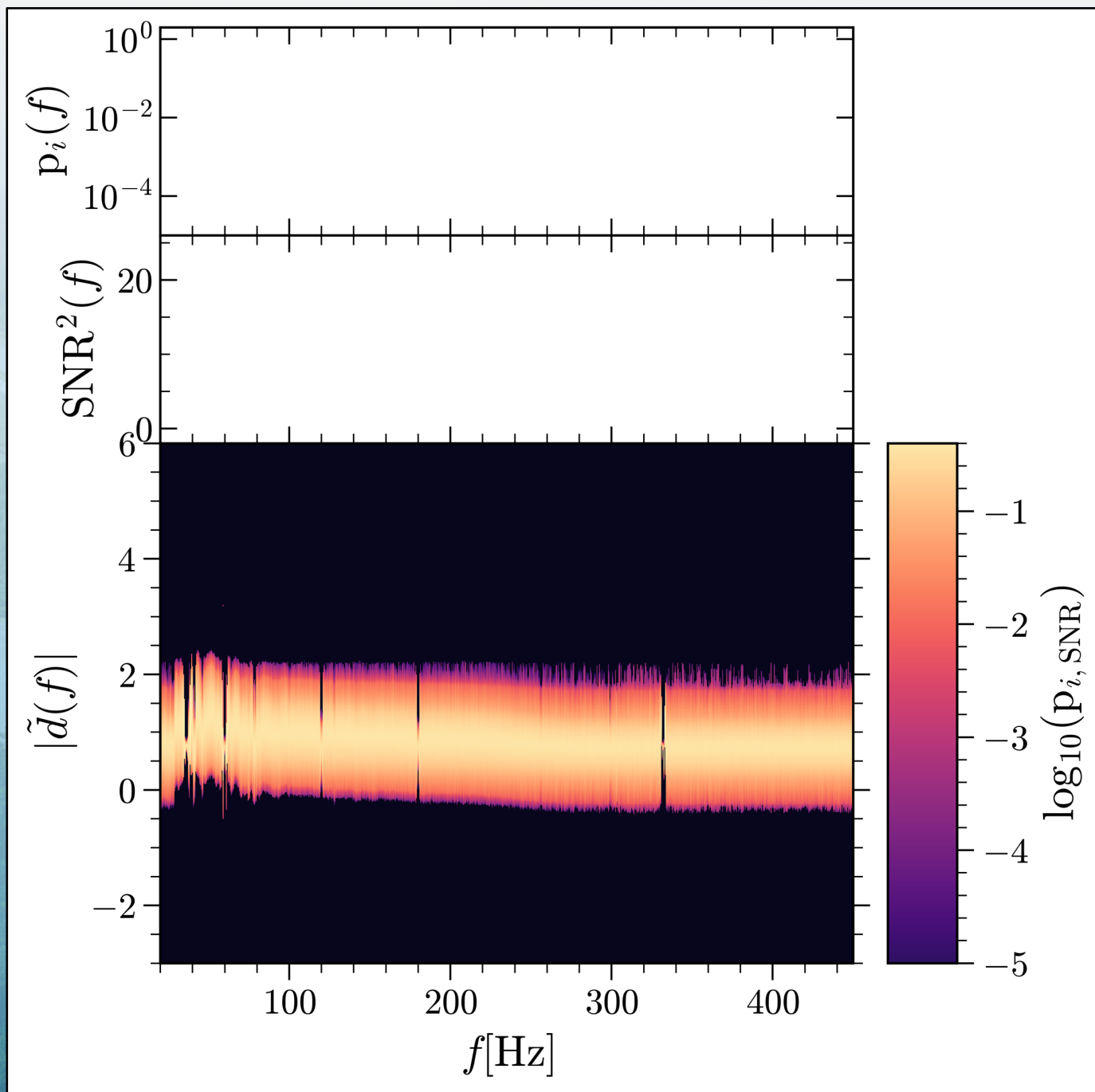
$$P^2 = \frac{4\pi^2}{GM} a^3$$

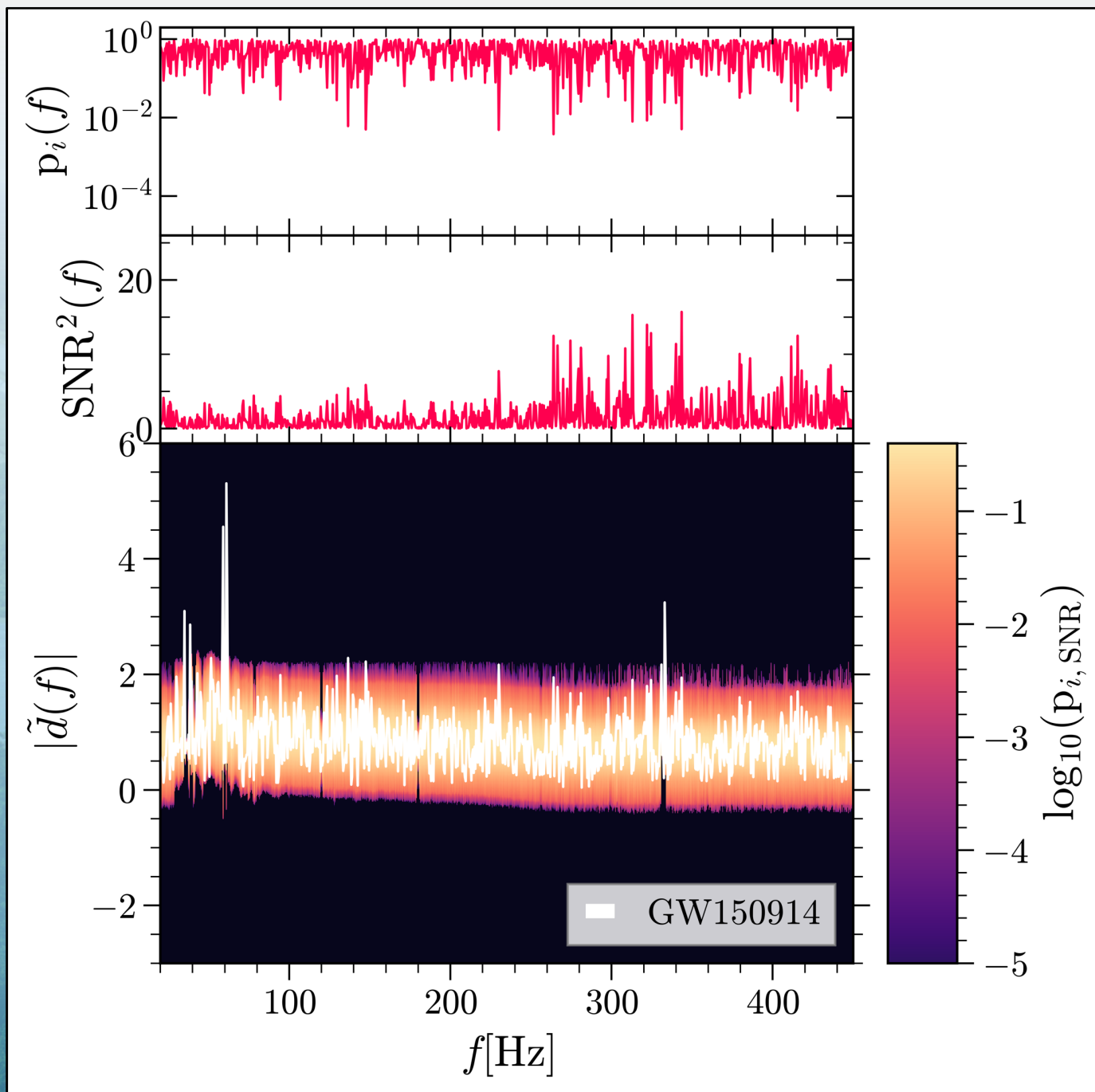
$$\Delta\phi \approx \frac{6\pi GM}{c^2 a (1 - e^2)}$$

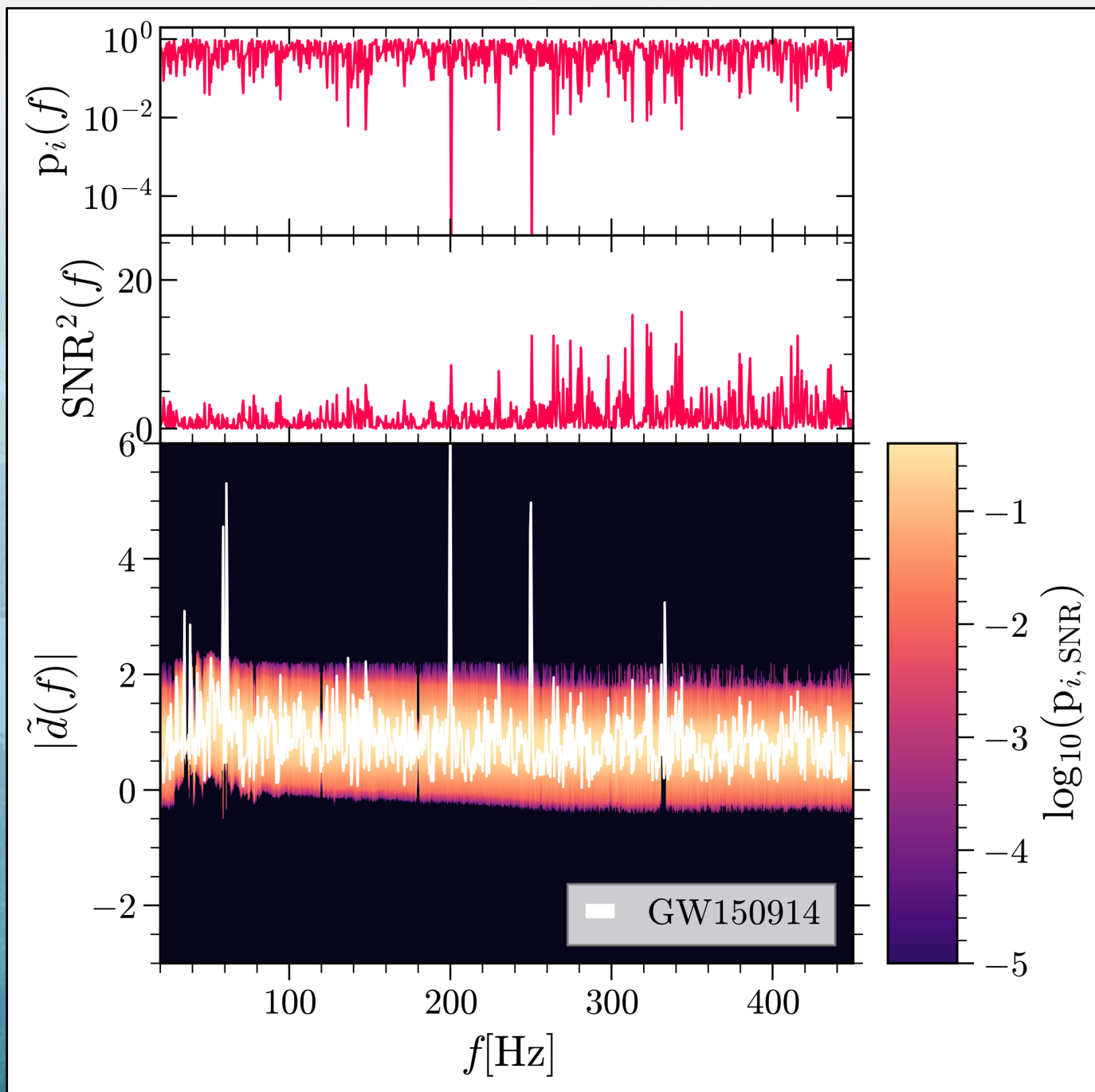


?

7







Does the Model Fit the Data?

*Machine learning tools to detect gravitational wave mismodelling
(Summer '25 @ the IoA)*

PHYSICAL REVIEW D 111, 083013 (2025)

Tests for model misspecification in simulation-based inference: From local distortions to global model checks

Noemi Anau Montel^{1,*}, James Alvey^{2,3,†} and Christoph Weniger^{4,‡}

¹Max-Planck-Institut für Astrophysik, Karl-Schwarzschild-Strasse 1, 85748 Garching, Germany

²Kavli Institute for Cosmology Cambridge, Madingley Road, Cambridge CB3 0HA, United Kingdom

³Institute of Astronomy, University of Cambridge, Madingley Road,

Cambridge CB3 0HA, United Kingdom

⁴GRAPPA Institute, Institute for Theoretical Physics Amsterdam, University of Amsterdam, Science Park 904, 1098 XH Amsterdam, The Netherlands

Ⓜ (Received 8 January 2025; accepted 10 March 2025; published 7 April 2025)

Model misspecification analysis strategies, such as anomaly detection, model validation, and model comparison are a key component of scientific model development. Over the last few years, there has been a rapid rise in the use of simulation-based inference (SBI) techniques for Bayesian parameter estimation, applied to increasingly complex forward models. To move toward fully simulation-based analysis pipelines, however, there is an urgent need for a comprehensive simulation-based framework for model misspecification analysis. In this work, we provide a solid and flexible foundation for a wide range of model discrepancy analysis tasks, using *distortion-driven model misspecification tests*. From a theoretical perspective, we introduce the statistical framework built around performing many hypothesis tests for distortions of the simulation model. We also make explicit analytic connections to classical techniques: anomaly detection, model validation, and goodness-of-fit residual analysis. Furthermore, we introduce an efficient self-calibrating training algorithm that is useful for practitioners. We demonstrate the performance of the framework in multiple scenarios, making the connection to classical results where they are valid. Finally, we show how to conduct such a distortion-driven model misspecification test for real gravitational wave data, specifically on the event GW150914.

DOR: 10.1103/PhysRevD.111.083013

I. INTRODUCTION

The primary goal of the physical sciences is to refine analytic or computational models that form the backbone of our understanding of physical phenomena. Key steps in this process are designing, fitting, and validating different models against data. There are a variety of existing strategies to approach this. Powerful tools include Bayesian evidence estimation conditioned on observations for model comparison [1], or goodness-of-fit and hypothesis tests for establishing the validity of a model across the observed data and making discoveries [2,3].

In recent years, simulation-based inference (SBI), also known as implicit-likelihood inference, has emerged as an important tool for inference when simulations from implicitly defined models are available [4]. This approach is especially useful for managing the increasing dimensionality of datasets and the growing complexity of scientific models, where often a full probabilistic description is difficult or even impossible to define. It is also useful in the regime where classical methods are computationally demanding, but efficient simulations remain feasible.

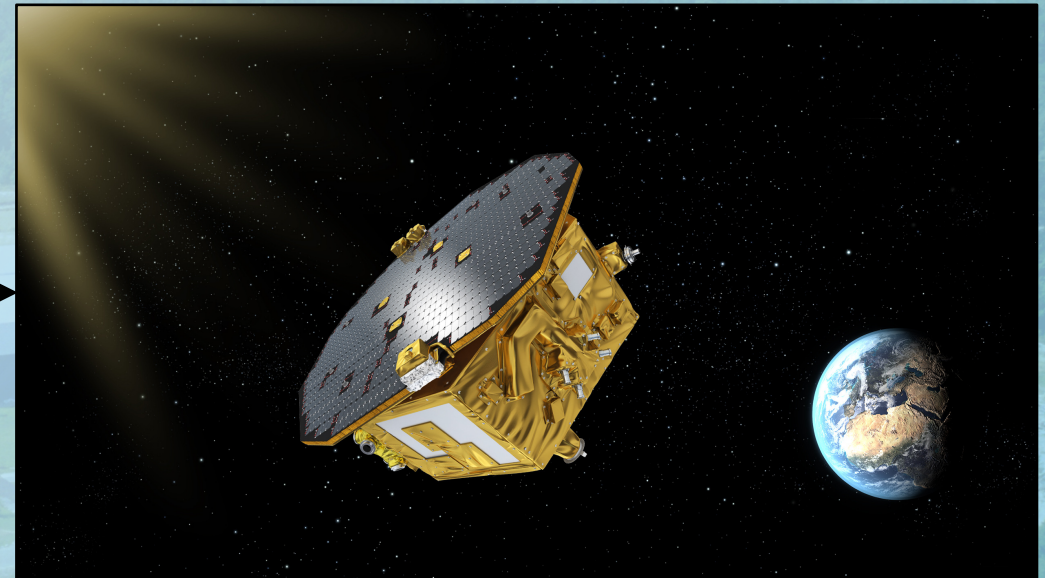
In SBI, the modeling complexity is shifted from having to define a likelihood function to having to program a simulator, which may be easier than constructing an analytical probabilistic description. In principle, this allows one to include and account for more effects in the modeling than in traditional methods. However, as in any statistical inference framework, the question of model misspecification—how to detect it, how it affects parameter reconstruction, and how to account for it—remains. In the SBI setting, model misspecification occurs when the data-generating function (the simulator) is a poor representation of the physical processes being analyzed. Modeling choices in the simulator are thus extremely important and the degree

*Contact author: noemiam@mpa-garching.mpg.de

†Contact author: jba2@cam.ac.uk

‡Contact author: c.weniger@uva.nl

Published by the American Physical Society under the terms of the Creative Commons Attribution 4.0 International license. Further distribution of this work must maintain attribution to the author(s) and the published article's title, journal citation, and DOI. Open access publication funded by the Max Planck Society.



T G Hardy

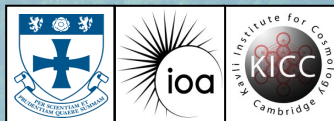
7th Nov @ IoA

Supervised by Dr James Alvey
KICC/IoA @ Cambridge University



Does the Model Fit the Data?

*Machine learning tools to detect
gravitational wave mismodelling
(Summer '25 @ the IoA)*



T G Hardy

7th Nov @ IoA

Supervised by Dr James Alvey
KICC/IoA @ Cambridge University

